

# Locus: An Offline-First Embedded Knowledge Agent Framework Using Geographic Coordinate Addressing

Daniel Ray  
Independent Researcher  
antfriend@gmail.com  
<https://github.com/antfriend>

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## Abstract

We present **Locus**, an offline-first embedded knowledge agent framework for resource-constrained microcontrollers (ESP32-class) and up. Locus introduces **@LATxLON** geographic coordinate addressing—an authority-free scheme in which every knowledge node is identified by a deterministic latitude/longitude pair requiring no registry, no network round-trip, and no external authority. Knowledge is stored as **toot-bits**, the minimal unit of semantic meaning, organised in **TTDB** (MyMentalPalaceDB): a flat Markdown or  $\text{\LaTeX}$  knowledge graph traversable with no database engine. Each TTDB instance is bound to an **umwelt**—a formally specified subjective worldview that gives a device its identity and reasoning perspective. The framework is governed by a multi-RFC specification suite covering the mesh network protocol (TTN), the TTDB file format (cursor semantics, typed edges, epistemic weight tracking), engineering workflow (TTE), and the ESP32 autonomous device framework (A32). Agents operate via deterministic sense-reason-act loops over TTDB graph traversal—no cloud LLM inference required. We provide a 21-node affective knowledge graph grounded in hippocampal-prefrontal emotional embedding research, a six-node medical knowledge graph integrating UMLS Concept Unique Identifiers, and a complete open-source reference implementation. All specifications, artifacts, and firmware are publicly available.

## 1 Introduction

Knowledge agents deployed at the edge face a fundamental tension: the richer the knowledge representation, the heavier the runtime dependency. Existing approaches either require cloud inference—introducing latency, connectivity dependency, and data sovereignty concerns—or strip semantics down to lookup tables, sacrificing expressiveness. Locus resolves this tension by separating the *knowledge format* from the *reasoning engine* and grounding both in a novel addressing scheme that requires no central authority.

The **@LATxLON** scheme assigns every knowledge record a coordinate-pair identifier derived from its position on a conceptual globe defined per-umwelt. Unlike URI schemes, which require an authority registry, or UUID schemes, which are opaque to humans, @LATxLON identifiers are deterministic, mnemonic, human-readable, and valid with zero network connectivity. A record at **@LAT30LON-20** is unambiguously located on its umwelt’s globe—traversable, linkable, and reproducible from the specification alone.

**TTDB** (MyMentalPalaceDB) stores knowledge as plain Markdown or  $\text{\LaTeX}$ , parsed with a two-pass indexed reader requiring approximately 12 bytes per record in RAM. **Agent 32 (A32)** runs

this parser on ESP32-S3 hardware (8 MB flash, 2 MB PSRAM), executing a 1-second sense-reason-act cycle entirely from local storage. Locus unifies these components under a formal multi-RFC specification suite, enabling reproducibility, community extension, and cross-device interoperability without cloud infrastructure.

## 2 Key Contributions

- @LATxLON addressing.** An authority-free, deterministic, human-readable namespace for knowledge nodes. Functions offline with no registration step. Scales from single-device to mesh-networked deployments.
- TTDB file format.** A flat Markdown/L<sup>A</sup>T<sub>E</sub>X knowledge graph specification with formal cursor semantics (TTDB-RFC-0002), typed directional edges (TTDB-RFC-0003), and toot-bit epistemic weight tracking (TBEW, TTDB-RFC-0005). No database engine required; parseable on a \$5 microcontroller.
- Toot-Bit Epistemic Weight (TBEW).** A per-record metadata block tracking confidence (`conf`, 0–255), revision count (`rev`), salience (`sal`), and last-touch timestamp (`touched`). The derived signal

$$\text{EPS} = \text{sal} \times \frac{255 - \text{conf}}{255}$$

identifies records that are heavily used but poorly understood—the primary attention targets for knowledge refinement.

- Umwelt formalism.** Operationalisation of Uexküll’s biosemiotic umwelt concept as a firmware-level device identity [1]. Two physically identical ESP32 devices loaded with different TTDB files exhibit different behaviours and reasoning paths—they inhabit different umwelten.
- Multi-RFC specification suite.** Formal specifications covering TTN (semantic mesh network protocol with trust, compression, and LoRa framing), TTDB (file format, cursor, typed edges, event IDs, TBEW, experiential perception), TTE (engineering workflow), and A32 (ESP32 firmware architecture, TTDB parsing, agent loop, hardware abstraction). Enables reproducibility and community fork/extend workflows.
- Affective knowledge graph.** A 21-node TTDB instance mapping feelings, emotions, dispositions, and intents on a valence/arousal coordinate space, grounded in Ma and Kragel [2]. Deployable to ESP32 as a standalone affective reasoning module.
- Medical knowledge graph.** A six-node TTDB storyline template for clinical knowledge representation: symptom → condition → molecule → medicine → perceptual experience → outcome. Integrates UMLS Concept Unique Identifiers [4] as a grounding layer for node identity.

## 3 System Overview

The Locus stack has three tiers.

**Storage tier (TTDB).** A flat text file (Markdown or  $\text{\LaTeX}$ ) containing a YAML `mmpdb` header, a `cursor` block, and record sections separated by horizontal rules. A minimal record takes the form:

```
@LAT30LON-20 | created:1778000000 | updated:1778200000
               | relates:supports>@LAT0LON0
```

Body text describing this knowledge node.

Each record carries a typed edge list, an optional `[ew]` epistemic weight block, and a free-text body.

**Agent tier (A32).** Firmware for ESP32-class hardware. At boot, the two-pass parser builds an in-memory index (12 bytes/record). The agent loop reads sensor inputs, quantises them to TTDB coordinates, finds the nearest record in the index, follows typed edges (`triggers`, `navigates_to`, `inhibits`, `requires`, `logs`), and drives actuators. Memory budget:  $\sim 3$  KB overhead + 12 bytes/record.

**Network tier (TTN).** An optional LoRa mesh network layer for multi-device deployments. Nodes communicate via semantic events rather than raw messages. A32 devices using LoRa implement TTN-RFC-0006 packet framing. Semantic compression tokens (`P`, `S?`, `OK`, `T:x`, ...) are gateway-expanded to full semantic events.

## 4 Related Work

**Knowledge graphs on embedded hardware.** Prior work (e.g., RDF-Turtle on Raspberry Pi-class devices) requires a SPARQL engine. TTDB achieves comparable expressiveness with a 3 KB parser suitable for bare-metal ESP32.

**Authority-free naming.** IPFS content addressing is hash-based and opaque. Decentralised Identifiers (DIDs) require a resolver. `@LATxLON` is position-based, human-readable, and resolver-free.

**Umwelt in robotics.** Several papers apply *umwelt* metaphors to robotic sensor modelling; none formalise it as a firmware-level identity specification with a flat-file knowledge graph.

**Offline AI agents.** TinyML frameworks (TensorFlow Lite Micro, Edge Impulse) use neural inference. Locus uses symbolic graph traversal—no training data, no model weights, fully inspectable.

## 5 Availability

All specifications, reference firmware, and TTDB collections are available under open licences:

- RFC suite: <https://github.com/toot-toot-engineering>
- TTDB collections and mega librarian: <https://antfriend.github.io>
- Zenodo dataset deposit: DOI to be confirmed

## References

- [1] J. von Uexküll, *Umwelt und Innenwelt der Tiere*. Gustav Fischer, 1909.

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- [3] K. Friston, “The free-energy principle: a unified brain theory?” *Nature Reviews Neuroscience*, vol. 11, no. 2, pp. 127–138, 2010.
- [4] O. Bodenreider, “The Unified Medical Language System (UMLS): integrating biomedical terminology,” *Nucleic Acids Research*, vol. 32, no. suppl\_1, pp. D267–D270, 2004.